CCIR · COMPUTE CREDIT INDEX RESEARCH

The Reference Standard for GPU Collateral.

Methodology Statement | Version 1.0.0 | March 2026

Item	Detail
Scope	Five spot indices (CRI-H100, CRI-A100, CRI-H200, CRI-L40S, CRI-B200) and cluster indices at two tiers (CLS-MM, CLS-ENT) for eligible GPU models
Primary use cases	Collateral coverage / LTV covenant floors, maintenance test anchors, generation obsolescence monitoring via cross-model spread
Primary rate	On-demand spot rate; available without preemption risk. Published as the primary CRI headline rate (seller_ask_noninterruptible).
Computation	Flat median (percentile_approx 0.5) across all Silver-eligible observations in a trailing 7-day window
Independence	CCIR holds no financial interest in any GPU marketplace from which it collects data — a structural prerequisite for credit document citation
Methodology version	v1.0.0 — stamped on every Bronze, Silver, and Gold row; changes follow the published Benchmark Change Management and Versioning Policy

All five spot indices carry multi-source coverage on the primary published rate (seller_ask_noninterruptible: RUNPOD, SHADEFORM, LAMBDA, TENSORDOCK, NEBIUS — source mix varies by model; see Table 2). No spot index carries structural low confidence on the primary rate. Structural low confidence applies to CLS-ENT (single enterprise source by design) and to CLS-MM for H200 (Lambda does not carry H200 in its 8x cluster offering). The interruptible rate (seller_ask_interruptible) is collected and stored but not published as a headline index value. No structural low confidence designation applies to any published spot index. Section 4 explains confidence flag interpretation.

This document describes the methodology governing CCIR's published reference rates for GPU compute. It covers eligibility criteria, computation procedures, confidence flag definitions, the spread framework, and source taxonomy. All design decisions are documented with the rationale required to assess fitness for use in credit documents, collateral appraisals, and covenant frameworks.

CCIR holds no financial interest in any GPU marketplace from which it collects data. This structural independence is a prerequisite for use of CCIR rates as reference instruments in lending agreements. The methodology reflects that position: every filter and computation choice is made to maximize representativeness and auditability, not to favor any particular market participant or price level.

1. Reference Instrument Definitions

Each CRI series is an independent reference rate for a specific GPU collateral type. No single floor rate exists from which others derive. A lender underwriting H100 collateral cites CRI-H100; L40S collateral cites CRI-L40S. This structure mirrors the AVITAS model for aircraft appraisals, where each aircraft type is valued independently.

1.1 Spot Rate Indices

The primary headline rate for each index is the seller-posted non-interruptible spot price: the on-demand rate at which compute is available without preemption risk, reflecting the stable floor for covenant maintenance testing. The interruptible rate is also collected but not published as a headline index value.

Index	GPU Model	Form Factor	Primary Rate	Sources (Noninterruptible)
CRI-H100	NVIDIA H100 80GB	SXM (NVLink)	seller_ask_noninterruptible	RUNPOD, SHADEFORM, LAMBDA, TENSORDOCK, NEBIUS
CRI-A100	NVIDIA A100 80GB	SXM4 (NVLink)	seller_ask_noninterruptible	RUNPOD, SHADEFORM
CRI-H200	NVIDIA H200	SXM (NVLink)	seller_ask_noninterruptible	RUNPOD, SHADEFORM, NEBIUS
CRI-L40S	NVIDIA L40S	PCIe	seller_ask_noninterruptible	RUNPOD, SHADEFORM, TENSORDOCK, NEBIUS
CRI-B200	NVIDIA B200	SXM (NVLink)	seller_ask_noninterruptible	RUNPOD, SHADEFORM, LAMBDA, NEBIUS

All five spot indices are multi-source on the noninterruptible side. Five sources contribute seller_ask_noninterruptible observations: RUNPOD (securePrice), SHADEFORM, LAMBDA, TENSORDOCK, and NEBIUS. Source coverage varies by model — see Table 2 for per-index source detail. VAST and RUNPOD minimumBidPrice contribute seller_ask_interruptible observations. These are collected and stored but not published as headline index values. Source count and confidence flag are published daily alongside each index value.

1.2 Cluster Indices

Cluster indices represent the cost of coordinated multi-GPU compute at two tiers: mid-market (MM) and enterprise (ENT). They are published alongside the corresponding spot rate to support spread analysis and covenant structuring.

Index	Tier	GPU Model	Sources
CRI-H100-CLS-MM	MM	H100 SXM	RUNPOD_CLUSTER, LAMBDA_CLUSTER
CRI-A100-CLS-MM	MM	A100 SXM	RUNPOD_CLUSTER, LAMBDA_CLUSTER
CRI-B200-CLS-MM	MM	B200 SXM	RUNPOD_CLUSTER, LAMBDA_CLUSTER
CRI-H100-CLS-ENT	ENT	H100 SXM	COREWEAVE
CRI-H200-CLS-ENT	ENT	H200 SXM	COREWEAVE
CRI-B200-CLS-ENT	ENT	B200 SXM	COREWEAVE

1.3 Canonical Instrument Definition

> The CRI-H100 index is the median seller-posted non-interruptible on-demand price for one hour of a single NVIDIA H100 SXM GPU, drawn from executable listings across GPU compute marketplaces, subject to full-GPU allocation, minimum host reliability, and one-signal-per-decision-maker deduplication filters, computed as a simple flat median across all qualifying observations in the trailing 7-day window.

The definition above applies by substitution to all spot indices. Cluster indices substitute the per-GPU median rate for a coordinated 8-GPU pod, with source and bundle disclosures as specified in Section 5.

2. Eligibility Criteria

Bronze collects all available listings from each source without filtering. Eligibility filters are applied exclusively at the Silver layer. This design ensures that every filter decision is auditable against the raw Bronze data and that a methodology change does not require re-collection.

Silver filters enforce economic comparability: every row reaching Gold represents one executable, single-GPU, per-hour, seller-posted ask price. The filters below are locked as of methodology version v1.0.0. Any change triggers a version bump.

CCIR reserves the right to exclude data that is demonstrably erroneous or non-representative of executable market conditions, acting in good faith and in a commercially reasonable manner, provided that any such exclusion is documented in the methodology changelog with the rationale and the specific records excluded. Exclusions applied systematically to a class of observations constitute a methodology change subject to the versioning and notice requirements of the Benchmark Change Management and Versioning Policy. Ad hoc exclusion of individual records that are clearly erroneous (e.g., data transmission errors, obvious API anomalies) may be applied as a patch-level correction without advance notice.

2.1 One Signal Per Decision-Maker

One price signal per economic decision-maker per snapshot. A provider's decision to offer compute at a given price is a single market observation, regardless of how many regions, block sizes, or CPU variants that provider deploys at that price. This principle governs all source-specific deduplication rules in Sections 2.2 through 2.8.

Regions: Geographic deployment locations offered by a single provider for the same GPU model. Regional granularity reflects a provider's operational footprint — where they have built or leased datacenter capacity. A provider deploying H100s across five regions at $3.34/hr has made one pricing decision, not five. Where a provider posts distinct prices by region, the median across that provider's regions is taken as the single observation, preserving the provider's central tendency while maintaining the one-signal principle. Median is used rather than the lowest or highest regional price because a liquidating lender cannot predetermine which region the collateral will clear in — the median across the provider's available regions is the most neutral and auditable estimate of realizable price across the full deployment footprint. Regional price detail remains available in Bronze for drill-down analysis.

Block sizes: Minimum GPU allocation tiers (1×, 2×, 4×, 8× GPU) offered by a marketplace host for the same physical server. Block-size listings reflect inventory management and bundle economics, not independent pricing decisions. The Vast.ai machine_id deduplication implements this principle: one physical server = one observation per snapshot, with median price taken across block sizes.

CPU variants: Different CPU platform pairings offered alongside the same GPU model at the same price point. CPU platform selection is an infrastructure configuration choice by the provider, not a GPU pricing decision. The Nebius L40S is the current instance: Intel Ice Lake and AMD EPYC Genoa variants are offered at identical GPU component prices. One observation per (snapshot_hash, gpu_model, price_type_canonical) is taken.

2.2 Vast.ai

- gpu_frac ≥ 1.0 — Full GPU allocations only. Fractional pricing carries an above-linear premium not representative of production-scale deployment rates.
- is_bid = False — Seller-posted prices only. Buyer bids are not executable seller offers.
- reliability2 ≥ 0.90 — Excludes hosts with poor uptime history. Datacenter-grade reliability threshold.
- machine_id deduplication — One physical server may appear as multiple block-size listings. Rows are grouped by (snapshot_hash, gpu_model, machine_id) and the median price is taken across block sizes. One server = one observation per snapshot.
- hosting_type — not applied. The majority of Vast.ai supply is hosting_type=0; these listings are valid market observations. gpu_frac ≥ 1.0 is the primary comparability gate.

2.3 RunPod

- secure_cloud = True — Secure Cloud (datacenter-grade) only. Community Cloud excluded: consumer and prosumer hardware does not meet datacenter-grade comparability criteria.

RunPod returns aggregated market prices per GPU type, not individual machine listings. No machine_id or gpu_frac deduplication required. Each collection run produces up to two rows per GPU model (interruptible and non-interruptible).

minimumBidPrice is tagged as the interruptible spot bid floor — the price at which RunPod hosts will accept interruptible deployments. This is an executable price and a legitimate floor observation. securePrice is tagged as the non-interruptible on-demand rate and is the primary index input.

2.4 Shadeform

- available = True — Only regions currently accepting deployments. Unavailable listings are not current market prices.
- No datacenter quality gate — Shadeform does not expose a provider tier or reliability score in its API. The available = True filter is an availability gate only. All other CRI sources carry an explicit datacenter-grade criterion (RunPod: secure_cloud = True; Vast.ai: reliability2 ≥ 0.90; TensorDock: tier ≥ 3). Shadeform providers active in the CRI pipeline as of v1.0.0 (DigitalOcean, Hyperstack, Scaleway) are datacenter-grade infrastructure operators. This characteristic is disclosed rather than treated as a disqualifying filter gap.
- Provider-level deduplication — Shadeform is a multi-provider aggregator; a single collection run may return multiple provider/region combinations for the same GPU model. Each underlying cloud provider (identified by the cloud field) is the economic decision-maker — it sets its own price independently of other providers. Rows are grouped by (snapshot_hash, cloud, gpu_model, price_type_canonical) and the median price is taken across that provider's available regions. One provider = one observation per snapshot. This implements the one-signal-per-decision-maker principle (§2.1): regional deployment breadth does not constitute additional price signals. Where a provider posts distinct prices across regions, the median of those regional prices is the single observation. Regional price detail is preserved in Bronze for drill-down analysis. Providers collected directly via dedicated Bronze notebooks (Nebius, Lambda) are excluded from the Shadeform feed via the EXCLUDED_CLOUDS registry to prevent double-counting.

2.5 Lambda Labs

- capacity_available = True — Only when Lambda has live capacity. Sold-out rate cards are not executable prices.
- is_cluster = False — Single-GPU instances only. Multi-GPU cluster instances are handled by the cluster index pipeline.
- CRI-eligible models — H100_SXM, B200_SXM. A100_SXM excluded: Lambda's single-GPU A100 is 40 GB; CRI-A100 tracks 80 GB SXM4 only.

2.6 TensorDock

- available = True — max_count > 0; location accepting deployments.
- tier ≥ 3 — Datacenter-grade only. TensorDock tier 1/2 hosts are consumer/prosumer hardware. Tier 3/4 enforces ISO 27001 / datacenter-grade physical security consistent with collateral underwriting standards.

2.7 Nebius

- No filter required. Nebius Bronze is a posted rate card scraped from the public compute pricing docs page (docs.nebius.com/compute/resources/pricing). No API key required. All records are Silver-eligible.
- Models collected: H100_SXM, H200_SXM, B200_SXM, L40S. A100 not offered by Nebius — does not contribute to CRI-A100. Both on-demand (seller_ask_noninterruptible) and preemptible (seller_ask_interruptible) price types are collected. L40S is priced on a per-component basis by Nebius; the GPU component price only is collected, consistent with per-GPU-hr methodology.
- L40S CPU variant deduplication (§2.1) — Nebius offers L40S on two CPU platforms (Intel Ice Lake, AMD EPYC Genoa) at identical GPU component prices. Rows are grouped by (snapshot_hash, gpu_model, price_type_canonical) and one observation is taken. H100_SXM, H200_SXM, and B200_SXM each produce one row per price type per snapshot and are unaffected.

2.8 Geographic Scope

No geographic filter is applied. GPU compute is treated as a globally fungible asset: hardware can be redeployed across jurisdictions, and rental pricing across major datacenter markets reflects a connected global market. This treatment is consistent with collateral valuation in cross-border aircraft financing structures (EETC).

US-only observations are computed as a supplementary breakdown and reported in the Silver run summary. They are not used as a filter and do not affect published index values. Credit agreements referencing CCIR indices may specify a US-only rate variant if required by counsel; CCIR does not currently publish a US-only index series.

3. Computation

3.1 Rolling Window

Each daily index value is published following the evening pipeline run (typically 9:00 PM CT) and reflects the 7 completed calendar days ending on the day prior to publication. The as_of_date field stamps the publication date; the window_start and window_end fields record the exact observation window. The window rolls forward each calendar day. All Silver-eligible observations collected within the window contribute to the index, regardless of which day within the window they were collected. Because the window consists entirely of completed collection days, every source contributes its full daily observation depth with no partial-day asymmetry.

3.2 Aggregation Method — Flat Median

The index value is the simple median (50th percentile approximation) of all Silver-eligible price observations for a given (gpu_model, price_type_canonical) cell within the 7-day window. This is a flat median across all pooled observations — not a time-weighted average, not a median-of-medians, and not a median of daily medians.

The flat median is the transparent and appropriate approach for this application because Silver filters enforce economic comparability upstream. Every row reaching Gold represents one executable, single-GPU, per-hour, seller-posted ask. With apples-to-apples inputs, pooling all observations and taking a single median is the most transparent and least arbitrary computation available. A median-of-medians would silently exclude valid observations on days when fewer sources were active, creating artificial instability.

3.3 MERGE Idempotency

Gold is written via MERGE, keyed on (as_of_date, gpu_model, price_type_canonical) for spot indices and (as_of_date, gpu_model, cls_tier) for cluster indices. Re-running Gold for the same date updates the existing row rather than appending a duplicate. This makes each daily run idempotent and supports retroactive correction of upstream data without row proliferation.

3.4 Collection Schedule

Bronze collection runs four times daily at 06:00, 10:00, 14:00, and 18:00 CT. An additional full pipeline run (Bronze through export) executes at 21:00 CT. The 7-day window therefore accumulates up to five collection snapshots per day per source, increasing observation depth relative to a once-daily schedule.

4. Confidence Flags and Disclosure

Every published index value carries a confidence flag. No value is suppressed regardless of confidence level. Low-confidence values are published with explicit disclosure so that users can assess suitability for their specific use case.

Low confidence has two distinct causes. Transient low confidence occurs when data has not yet accumulated to meet the medium threshold — expected early in a new index's history and resolves over time. Structural

low confidence occurs when a source simply does not carry a given model on a given price type, making it impossible by market structure to achieve a second-source confirmation. Transient low warrants patience; structural low warrants consulting total_observations and source_count directly.

Flag	Source Threshold	Day Threshold	Interpretation
high	≥ 3 distinct sources	≥ 5 valid days	Multi-source, multi-day — highest reliability
medium	≥ 2 distinct sources	≥ 3 valid days	Two-source confirmation over most of the window
low (transient)	< 2 sources or < 3 valid days	New index or recovering source	Check total_observations; flag expected to improve
low (structural)	< 2 sources by market structure	Permanent	No second source exists for this model/price type — flag will not improve; assess via total_observations

n_valid_days is defined as the number of calendar days within the 7-day window on which at least 2 distinct sources each had at least one Silver-eligible observation for the model. Days on which only one source was active do not increment n_valid_days.

4.1 Structural Low Confidence — Current Cases

All five spot indices (seller_ask_noninterruptible) are multi-source on the primary published rate, with five contributing sources across models. No spot index carries structural low confidence on the primary rate.

The interruptible rate (seller_ask_interruptible) is collected and stored but not published as a headline index value. No structural low confidence designation applies to any published spot index.

The following cluster tiers carry structural low confidence by design:

- CLS-ENT (all models) — CoreWeave is the sole ENT source. source_count = 1 is expected and disclosed. ENT low confidence is an expected PASS in the nightly audit.
- CRI-H200-CLS-MM — Lambda does not carry H200 in its 8x cluster offering. RunPod is the sole MM source for this model.

Users assessing these indices for credit document use should examine total_observations and source_count directly. A single-source index with 50+ observations per window is meaningfully different from one with 3. The flag alone does not convey this distinction.

5. Spread Framework

CCIR publishes a family of reference rates. Index selection, chip weighting, and tier applicability are determined by the credit agreement — not by CCIR. A lender underwriting a mixed H100/A100 pool constructs a blended rate from CRI-H100 and CRI-A100 at deal-defined weights. CCIR's role is to publish the components with full methodology disclosure so that counsel and counterparties have the rate inputs required to structure and test covenants. This mirrors the AVITAS model: AVITAS publishes appraised values for each aircraft type; the EETC structurer selects which values apply to the specific collateral in the trust.

Three rate tiers are published per eligible GPU model. The horizontal spread between GPU models is the primary published signal for generation obsolescence and collateral impairment risk. The Generation Spread — CRI-H100 relative to CRI-B200 — measures the forward trajectory of collateral earning capacity as newer hardware reaches the market.

Tier	Field	Credit Function
Non-interruptible spot rate	seller_ask_noninterruptible	Primary reference rate (CRI-X headline). On-demand spot rate without preemption risk. Covenant maintenance test anchor.
Interruptible spot rate	seller_ask_interruptible	Collected and stored in the pipeline. Not published as a headline index value. Available for internal pipeline analysis.

Tier	Field	Credit Function
Mid-market cluster	cluster_ask_mm	Revenue-grade compute rate for ML training and inference workloads. CLS-MM.
Enterprise cluster	cluster_ask_ent	Enterprise on-demand cluster rate inclusive of bundled infrastructure. CLS-ENT.

5.1 Cross-Model Spread

Because CCIR publishes independent reference rates for each GPU model simultaneously, the spread between any two models is directly observable. The CRI-B200 / CRI-H100 spread measures the premium of next-generation hardware over the current workhorse generation. Compression of this spread — B200 prices falling toward H100 levels — signals that the training premium justifying H100 collateral valuations is eroding. The risk is not one-directional: H100 rates have shown resilience as inference demand absorbs capacity previously dedicated to training. The Generation Spread captures both dynamics in real time.

Cross-model spreads are available for use in covenant frameworks monitoring workload migration, generation obsolescence, and collateral substitution risk. The §7.04 Generation Spread covenant structure is built on this signal.

5.2 T2I Spread — Tier-to-Interruptible

The T2I Spread (Tier-to-Interruptible Spread) is computed internally as: (cls_rate − interruptible_rate) / cls_rate. The interruptible rate and T2I Spread are collected and retained in the pipeline for internal analysis and future methodology development. They are not published as headline index values. No T2I index series appears in any published CRI output file.

5.3 Cluster Index Methodology

5.3.1 CLS-MM (Mid-Market)

CLS-MM is the flat median across all Bronze rows from RunPod Cluster and Lambda Cluster within the 7-day window, for each eligible GPU model. The same median method applies as for spot indices: all observations are pooled and a single percentile_approx(0.5) is computed. CLS-MM is sourced from GPU rental marketplaces that publish per-GPU cluster pricing for multi-GPU pods. Prices reflect spot-market rental rates without committed-term discounts.

5.3.2 CLS-ENT (Enterprise)

CLS-ENT is the CoreWeave posted rate card, normalized to per-GPU-hour by dividing the 8-GPU HGX node price by 8. CoreWeave pricing is scraped from the public pricing page (coreweave.com/pricing). No API is required. The price is a posted rate card, not a market-clearing price.

CLS-ENT rates include bundled CPU (128 vCPUs), RAM (2,048 GB), local storage (61.44 TB), and InfiniBand interconnect. This bundle inclusion is disclosed in the bundle_note field and in all published output. Users comparing CLS-ENT to CLS-MM or spot rates should account for the bundled infrastructure premium.

CoreWeave is currently the sole ENT source. source_count = 1 for all ENT rows. All ENT values carry a low confidence flag with full disclosure. The ENT tier will be upgraded to medium or high confidence as additional enterprise cloud providers are added.

6. Source Taxonomy

The table below characterizes each data source. Honest source characterization is essential for users assessing representativeness: a posted rate card, a marketplace ask, and an aggregated market bid floor are different economic things. CCIR documents each distinction explicitly.

Source ID	Type	API / Collection	Price Nature	Indices Fed
VAST	GPU rental marketplace	Public REST API (no key required)	Seller-posted interruptible asks from individual host operators. Interruptible only.	All spot indices (interruptible rate collected but not published as headline index value)
RUNPOD	GPU rental marketplace	Authenticated GraphQL API	Aggregated market prices per GPU type. minimumBidPrice (interruptible floor); securePrice (non-interruptible).	All spot indices (interruptible floor); all spot indices (noninterruptible primary rate)
SHADEFORM	GPU marketplace aggregator. Providers collected directly (Nebius, Lambda) excluded via EXCLUDED_CLOUDS registry to prevent double-counting.	Authenticated REST API	Seller-posted non-interruptible asks aggregated from underlying cloud providers. Non-interruptible only.	All spot indices (noninterruptible primary rate)
LAMBDA	GPU cloud provider	Authenticated REST API	Published rate cards for live capacity. Non-interruptible on-demand. Excluded when sold out.	CRI-H100, CRI-B200 spot indices (when capacity available). Lambda Bronze also collects GH200 (Grace Hopper) — time-series purposes only, feeds no current CRI index. Lambda Cluster feeds CLS-MM (see LAMBDA_CLUSTER row).
TENSORDOCK	GPU VM marketplace	Authenticated REST API (Bearer token)	Seller-posted non-interruptible VM pricing. Single price tier — no interruptible tier available.	All spot indices (noninterruptible depth — H100_SXM, A100_SXM, L40S)
RUNPOD_CLUSTER	GPU rental marketplace	Authenticated GraphQL API	Per-GPU-hr median cluster pricing for multi-GPU pods. Marketplace rates, no term commitment.	CLS-MM indices
LAMBDA_CLUSTER	GPU cloud provider	Authenticated REST API	Published 8x GPU cluster rate cards. Per-GPU-hr normalized.	CLS-MM indices (H100, B200, A100)

Source ID	Type	API / Collection	Price Nature	Indices Fed
COREWEAVE	Enterprise GPU cloud	Public web scrape (no key required)	Posted rate card for 8-GPU HGX nodes. Per-GPU normalized (node price / 8). Includes bundled CPU/RAM/storage/interconnect.	CLS-ENT indices (H100, H200, B200)
NEBIUS	GPU cloud provider	Public web scrape (no key required) — docs.nebius.com/compute/resources/pricing	Posted rate cards for on-demand and preemptible per-GPU-hr pricing. Both price types collected. No normalization required — prices already expressed per-GPU-hr. No availability state to filter — all records are Silver-eligible.	All spot indices where Nebius carries the model: CRI-H100, CRI-H200, CRI-B200 (noninterruptible + interruptible); CRI-L40S (noninterruptible + interruptible). Does not carry A100.

6.1 Audit Trail

Every Bronze row carries a SHA-256 snapshot hash computed from the raw API response for that collection run. The hash links each Silver and Gold value to the specific raw collection event that generated it. Bronze tables are append-only and never modified after write. Filter decisions are applied exclusively at the Silver layer, making every eligibility decision auditable against the underlying Bronze data.

Methodology version is stamped on every Bronze, Silver, and Gold row. Any change to a filter threshold, computation method, or source inclusion triggers a version bump, maintaining a full record of which methodology governed each published value.

7. Governance and Independence

CCIR is structurally independent from all GPU compute marketplaces. It holds no financial interest — equity, revenue share, referral fee, or other arrangement — in any source from which it collects data. This independence is not rhetorical: it is the structural precondition for CCIR indices to serve as reference instruments in credit documents without creating the conflicts of interest that have disqualified lender-controlled metrics in post-LIBOR structured finance practice.

Methodology changes require documentation of the rationale, a version bump, and prospective application only. Historical values computed under a prior methodology version are not restated.

- Oversight Board formation: targeted for Q1 2027
- Annual independent audit: targeted for first full calendar year of publication

7.1 Monthly Review Obligation

GPU API naming conventions change without notice. A sudden drop in observation count for any model is the primary indicator of a gpu_name string change on a source platform. Source-specific model variant registries are reviewed monthly against Bronze gpu_model_raw values. Any mismatch is corrected and documented in the methodology changelog.

All audit results are written to the audit/ Delta tables in ADLS Gen2, keyed on (audit_date, check_name, audit_source). This provides a persistent, queryable record of pipeline health independent of job logs.

8. Data Quality Controls

CCIR operates six automated audit notebooks that run independently from the data collection and index computation pipeline. Results are written to the audit/ Delta tables in ADLS Gen2 and surface as PASS, WARN, or FAIL. Any FAIL raises an exception that triggers a Databricks job failure notification.

Job	Schedule	Scope	Failure Behavior
audit_pipeline	Nightly — runs as final task in the Evening Pipeline after export_to_github	Reads published GitHub CSVs end-to-end. Validates column counts, today's row presence, sanity bounds, observation floors, source counts, and confidence flags.	Raises on FAIL → job failure notification. WARNs logged but do not block.
audit_sources	Weekly — Sunday 07:00 CT, separate job	API auth checks, required field presence, schema diffs against ADLS snapshots, and new GPU model string detection for each source.	WARNs expected (new GPU candidates, schema drift). Does not raise — weekly human review required.
layer_reconciliation	Nightly — runs in parallel with audit_pipeline after export_to_github	Row count reconciliation across Bronze → Silver → Gold using Delta transaction logs. Per-source Silver contribution check. FAIL on zero Gold rows or unreadable Delta history.	Raises on FAIL. WARNs logged (e.g. Lambda sold out, source at zero obs) but do not block.
delivery_manifest	Nightly — runs in parallel with audit_pipeline after export_to_github	Per-publication SHA-256 delivery manifest. Fetches all 11 published CSVs, computes checksums, re-fetches and verifies. FAIL on checksum mismatch, missing file, or stale date.	Raises on FAIL. WARNs logged (unparseable rate) but do not block.
audit_digest_daily	Nightly — final task in Evening Pipeline after audit_pipeline, layer_reconciliation, delivery_manifest	Consolidated HTML email digest to research@ccir.io covering all nightly audit results: index values, file verification, layer reconciliation, and all audit_pipeline check results.	Does not raise. Delivery mechanism only.
audit_digest_weekly	Sunday — final task in audit_sources job	7-day HTML summary email to research@ccir.io: per-day audit summary, index value trend, audit_sources health results, layer reconciliation rollup, and file delivery summary.	Does not raise. Delivery mechanism only.

8.1 Nightly Pipeline Audit — Check Sets

audit_pipeline validates the following for each index file on each run:

- Column count — spot files must have 15 columns; cluster files must have 18 columns
- Today's row presence — at least one row with today's as_of_date must exist in each file
- Methodology version — all today's rows must carry the current version string
- Index value sanity bounds — index_value and cls_rate must fall within per-index bounds (see table below)

- T2I spread bounds — t2i_spread must be in the range [0.01, 0.95]. Internal integrity check only; T2I Spread is not a published headline index value.
- Observation floor — total_observations must meet per-index minimums
- Source count floor — source_count must meet per-index minimums (ENT single-source is expected PASS at 1)
- Confidence flag — ENT low is expected PASS; spot or MM low triggers WARN
- Source coverage flag — source_coverage value (FULL / REDUCED / MINIMAL) present on every spot row; REDUCED or MINIMAL triggers WARN
- Observation inflation ratio — today's observation count vs. 30-day rolling average per model; ratio >1.5× triggers WARN (possible double-ingestion signal)

8.2 Sanity Bounds

Sanity bounds define the plausible range for each index value. A value outside these bounds triggers a FAIL and halts the pipeline notification chain.

Index	Min ($/GPU/hr)	Max ($/GPU/hr)
CRI-H100	$0.50	$12.00
CRI-H200	$0.80	$20.00
CRI-A100	$0.20	$8.00
CRI-L40S	$0.20	$6.00
CRI-B200	$2.00	$40.00
CRI-H100-CLS-MM	$1.00	$16.00
CRI-H100-CLS-ENT	$1.50	$22.00
CRI-H200-CLS-ENT	$2.00	$32.00
CRI-B200-CLS-ENT	$3.00	$50.00
T2I Spread (all)	0.01	0.95

8.3 Weekly Source Audit — Check Sets

audit_sources validates each data source independently on a weekly cadence. Checks are source-specific and reflect the exact field names and API structures used in the Bronze collection notebooks:

- Auth — API key accepted and returns a valid response
- Required fields — all fields the CCIR parsing logic depends on are present in the response
- Schema diff — current response keys are compared against a stored ADLS snapshot; new or removed fields trigger a WARN
- New GPU models — response model strings are compared against the source-specific known set; unrecognized strings trigger a WARN for monthly review
- Source-specific structural checks — e.g. RunPod markup ratio plausibility, CoreWeave page size (guards against truncated scrape), TensorDock tier distribution, Nebius page size (guards against docs page restructure)

Schema snapshots are stored in ADLS at utils/schema_snapshots/{source}_schema.json and updated whenever a schema diff is detected and confirmed as intentional. This creates a versioned record of each source API's response structure over time.

Glossary

Terms used in this document and in CCIR published output:

Term	Definition
Bronze	Raw collection layer. Append-only Delta table. All data as collected from source APIs, before any eligibility filtering. Never modified after write.
Silver	Filtered observation layer. Source-specific comparability filters applied; common column schema; all spot sources unioned. Each row represents one eligible price observation.
Gold	Index computation layer. Flat median computed across Silver-eligible observations in the 7-day window. Published as the CRI index value.
seller_ask_noninterruptible	Seller-posted price for non-interruptible spot compute — the buyer cannot be preempted by the host. Primary CRI headline rate. Covenant maintenance test anchor.
seller_ask_interruptible	Seller-posted price for interruptible spot compute — the buyer can be preempted by the host. The most commodity-like clearing price. Collected and stored in the pipeline; not published as a headline index value.
cluster_ask_mm	Mid-market cluster rate. Median of RunPod and Lambda 8-GPU posted prices. CLS-MM.
cluster_ask_ent	Enterprise cluster rate. CoreWeave on-demand rate card, normalized to per-GPU by dividing 8-GPU node price by 8. Includes bundled CPU/RAM/storage/interconnect. CLS-ENT.
CRI-X	CRI (Compute Rate Index) - Generic notation for any CCIR spot index (e.g., CRI-H100, CRI-B200). The X denotes the specific GPU model.
CLS-MM	Mid-market cluster index. See cluster_ask_mm.
CLS-ENT	Enterprise cluster index. See cluster_ask_ent.
T2I Spread	Tier-to-Interruptible Spread. Formula: (cls_rate − interruptible_rate) / cls_rate. Computed and retained internally. Not published as a headline index value. Available in the pipeline for internal analysis and future methodology development.
snapshot_hash	SHA-256 hash of the raw API response for a given collection run. Stamped on every Bronze row. Links Silver and Gold values to the specific collection event that generated them.
methodology_version	Semantic version string (e.g., v1.0.0) stamped on every Bronze, Silver, and Gold row. Any change to a filter, computation method, or source triggers a version bump.
n_valid_days	Number of calendar days in the 7-day window on which at least 2 distinct sources each had at least one Silver-eligible observation. Used to determine medium vs. low confidence flag.
confidence_flag	Published quality indicator: high (≥3 sources, ≥5 valid days), medium (≥2 sources, ≥3 valid days), low (all other cases). Low has two subtypes: transient (resolves with data) and structural (permanent market constraint). See Section 4.
source_coverage	Published field on every spot Gold row indicating whether the full expected source universe contributed to the index. FULL: all expected sources contributed. REDUCED: source_count < expected but ≥2. MINIMAL: source_count < 2. Expected source counts are methodology-locked per (gpu_model, price_type_canonical) in gold_spot_rates. See Data Source Quality Policy for publication hold thresholds.

MERGE key	The unique key used to upsert Gold rows. Spot: (as_of_date, gpu_model, price_type_canonical). Cluster: (as_of_date, gpu_model, cls_tier). Prevents duplicate rows on re-run.
bundle_note	Disclosure field on CLS-ENT rows noting that the per-GPU rate includes bundled CPU, RAM, storage, and interconnect from the CoreWeave node pricing.
as_of_date	The publication date for this Gold index value. The observation window covers the 7 completed calendar days ending on as_of_date minus 1 (i.e. window_start = as_of_date minus 7, window_end = as_of_date minus 1). The as_of_date and window bounds are stored explicitly on every Gold row for audit.

CCIR Compute Credit Index Research · ccir.io · This document describes methodology as implemented in pipeline version v1.0.0.